Predicting Heart Disease from Phone Survey

Project Proposal

CMPT 455 Data Mining Fall 2026

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Project Definition

Topic Motivation

The motivation for this project is to be able to predict those who have had heart disease from a questionnaire. This is worth investigating because if a questionnaire can be given, it can then determine who should see a doctor. This would be a cost effective and simple way to screen people at risk of heart disease who should then visit a doctor.

What Data Set

Heart Disease data set from kaggle. Originally a 300 variable phone survey was conducted on general health from the CDC. The kaggle dataset we are using is a subset of this survey with 40 variables that may affect heart disease.

Kaggle: Personal Key Indicators of Heart Disease

There are a few datasets in the link. We will be working with the updated 2022 version with missing values. There is a data set that is already cleaned of missing values. However, for this project we will be working with the raw data set allowing us to clean it ourselves.

Data Set statistics

The data set has 445k samples. 44% of the rows contain a missing value. However the variable with the input feature with most missing values is tetanus shot at 19%. Not all the missing values are concentrated in one input feature.The target variable will be 'HeartDisease', it is positive if 'HadAngina' or 'HadHeartAttack' is postive. 9% of the samples are positive for 'HeartDisease'. This is an unbalanced class.This leaves 38 variables to be candidate input features. Of these 32 are categorical while 6 are numerical. The numerical values are 'PhysicalHealthDays', 'MentalHealthDays', 'SleepHours', 'HeightInMeters', 'WeightInKilograms', 'BMI'. Something of curiosity here is that BMI is a calculation based on height and weight. Is having all 3 important or is BMI redundant if you have the other two variables? A future test in prepossessing perhaps.For the categorical variables some examples are age (grouped in buckets), sex, ethnicity, smoking, cigarettes along with various other generic medical questions.

Data Mining Tasks

A single decision tree could be used for a survey where each answer determines the next question. Also creating a random forest would be useful to see how it compares in performance to the single tree. We will also try a logistic regression; as it predicts the probability of the sample belonging to a class. We can then fine tune the threshold probability for classifying someone as having heart disease.

When testing our models since the distribution in the target variable is not evenly split measuring accuracy is not as important as recall. If we predicted no for every person the accuracy would be ∼91%. We are screening for potential heart disease. This makes recall important as false positives are not as big of a concern. We are predicting from a phone survey and not medical information. If flagged from the model the next process would be going to a medical professional to confirm the diagnosis.

Related Work

By reviewing existing literature, a strong foundation is established for using machine learning to detect cardiovascular disease (CVD) in patients and improve outcomes. This section reviews three recent works that provide different perspectives on this problem: an overview comparing machine learning against conventional methods by Wan et al. [1], a comparative study on preprocessing methods by Alsabhan and Alfadhly [2] and a comprehensive review by Kumar et al. [3].

Wan et al. [1] provide an overview of machine learning methods against conventional risk assessment tools arguing conventional tools are limited by cost, accessibility and generalisability. They outline the machine learning workflow in healthcare, highlighting the importance of data collection, preprocessing and training. By training multiple models (K-nearest neighbours, Naive Bayes, Logistic Regression...) on multiple publicly available datasets such as the Cleveland Heart Disease dataset (University of California, Irvine) and the Framingham Cardiovascular Disease Prediction dataset, each can be evaluated using their confusion matrix, F1 score derived from that matrix and area under the ROC curve. They highlight the usefulness of the F1 score in unbalanced datasets where accuracy can be misleading by balancing precision and recall.

Alsabhan and Alfadhly [2] focuses on the effects of data preprocessing when training various machine learning models. Using the aforementioned University California, Irvine dataset 12 models were trained:

> Logistics Regression, Naive Bayes, Support Vector Machine, Division Tree, Random Forest, Light Gradient Boosting Machine, CatBoost, K-Nearest Neighbors, Gradient Boosting Machine, AdaBoost, Linear Discriminant Analysis and Artificial Neural Network.

These models were trained without any feature scaling, with standardization, with min-max scaling, and with normalization. The metrics used for comparison were accuracy, precision, recall, F1 score, Cohen's Kappa, log loss, and results show that feature scaling is not universally beneficial, helping some models but harming others. Normalization was particularly detrimental to logistic regression, artificial neural networks, support vector machines, and K-nearest neighbors.

Kumar et al. [3] take a broader scope in their review of CVD detection by analyzing the whole machine learning pipeline. They highlight five themes: heart disease detection and diagnostics, ML models and algorithms, feature engineering and optimization, emerging technologies in healthcare, and applications of AI across diseases, and of interest to our project is feature engineering and optimization. They note that feature selection significantly impacts performance of their models by addressing data imbalance and feature redundancy.

They also noted that models can become more interpretable by performing a SHapley Additive exPlanations (SHAP) analysis making those models more useful in clinical settings. Additionally, an exploration is made into augmenting dataset by creating synthetic samples of the minority class which shows improvements in model accuracy but can result in overfitting if too much synthetic data is added.

While these papers establish the value of machine learning, they all use features gathered from clinical testing while our aim is to perform the same task using surveyable data. Nonethe-less, the literature thus informs the scope and methodology of this project by highlighting these areas of importance. First, machine learning can improve CVD detection but is heavily dependent on data quality and preprocessing. Second, class imbalance is a major problem in medical datasets and generating synthetic data is an option. Third, accuracy cannot be a sufficient evaluation metric since the classes are imbalanced and F1 scores are essential especially in a medical setting where false negatives have serious consequences. Fourth, feature engineering can play a large role in improving performance and explainability.

References

[1] Siming Wan, Feng Wan, and Xi jian Dai. Machine learning approaches for cardiovascular disease prediction: A review. *Archives of Cardiovascular Diseases*, 118(10):554–562, 2025.

[2] Waleed Alsabhan and Abdullah Alfadhly. Effectiveness of machine learning models in diagnosis of heart disease: a comparative study. *Scientific Reports*, 15(1):24568, Jul 2025.

[3] Raman Kumar, Sarvesh Garg, Rupinder Kaur, M. G. M. Johar, Sehijpal Singh, Soumya V. Menon, Pulkit Kumar, Ali Mohammed Hadi, Shams Abbass Hasson, and Jasmina Lozanović. A comprehensive review of machine learning for heart disease prediction: challenges, trends, ethical considerations, and future directions. *Frontiers in Artificial Intelligence*, 8, 2025.